

Cross-Test Report

Member 5's Upload APIs (US-70, US-71), Unit Level

Project	Upcomer, Sprint 1
Tester	Member 1
Code tested	branch <code>member5_after_tests</code> (PR #18): <code>main</code> at <code>0e1f7d2</code> plus the fixes below
Report commit	<code>73a8ae3</code>
Test run	2026-10-05 18:39 UTC, Vitest 3.2.7 on Node 24.15.0
Test file	<code>tests/unit/cross-test-member-5-uploads.test.ts</code>

Verdict

75 of 75 tests pass, covering 77 recorded API calls. US-70 and US-71 meet every key check in the Sprint 1 plan: valid uploads, invalid or corrupt files, storage success and failure, and the file can be opened after upload. The first run on `main` found one deviation from the plan and two weaknesses (Findings 1–3). All three are **fixed** on this branch, and the tests now assert the fixed behaviour. Finding 4 is by design.

Test group	Tests	Calls	Passed
A. POST <code>/api/admin/uploads</code> — access control	6	6	6/6
B. POST <code>/api/admin/uploads</code> — request handling	5	5	5/5
C. POST <code>/api/admin/uploads</code> — form fields	28	28	28/28
D. POST <code>/api/admin/uploads</code> — file checks	17	17	17/17
E. POST <code>/api/admin/uploads</code> — storage and record failures	6	7	6/6
F. Upload then GET <code>/files/:fileId</code> — file retrievable after upload	7	14	7/7
G. <code>getUploadOptions</code> — course list for the upload form	2	0	2/2
H. <code>validateUploadFile</code> and <code>parseUploadForm</code> — direct calls	4	0	4/4
Total	75	77	75/75

Check	Result
Cross-test file	75/75 pass
Whole unit suite	484/484 pass, 28 files
<code>npm run lint</code>	Pass
<code>npx tsc --noEmit</code>	Pass
<code>npm run build</code>	Pass
<code>npm run test:e2e</code>	28/28 pass, against a production build and a real PostgreSQL 18.4 database
<code>npm run reliability</code>	Not run

Method

The tests call the real route handlers. Only three things are faked:

- **The database** (`@/lib/db`), with an in-memory store: courses `course-a` (EECE 350, professor `prof-a`) and `course-b` (EECE 351, professor `prof-b`), and the term `term-fall`.
- **The session cookie** (`next/headers`), which selects the caller: no session, `admin-token`, `student-token`, `expired-admin-token` or `forged-token`.

- **unlink**, in test E-03 only, to force a cleanup failure.

Everything else is real code:

- `POST /api/admin/uploads`
- `lib/uploads.ts` and `lib/upload-rules.ts`
- Member 4's `lib/file-metadata.ts`
- Member 1's `getAdminUser` and session check
- the logger
- files written to a temporary storage folder
- Member 3's `GET /files/:fileId`, which section F uses to open each uploaded file again

Findings

Finding	On main (0e1f7d2)	Fix on this branch	Tests
1. Unknown professor	400 <code>professor_not_assigned</code> ; the plan says unknown course or professor returns 404	Fixed. Maps to 404 in <code>STATUS</code> in <code>lib/uploads.ts</code> . Only uploads change; the US-72 edit page keeps its own handling.	C-18, C-19
2. ZIP renamed to .docx/.pptx	Accepted (201) on the 4-byte ZIP signature alone	Fixed. A DOCX must also contain [Content_Types].xml and word/, a PPTX [Content_Types].xml and ppt/. Otherwise 400 <code>content_mismatch</code> .	D-16, D-17
3. No Content-Length header	The whole body was parsed in memory before the 20 MB check	Fixed. The route streams the body, stops at 20 MB + 1 MB, cancels the stream and returns 413 .	B-05, D-10

Shared files touched by the fixes:

- `tests/unit/uploads.test.ts` (Member 5's tests): the DOCX/PPTX fixtures now contain the Office entry names, the professor case expects 404, and two corrupt-content cases were added.
- `tests/reliability/features/admin.ts`: the "professor not teaching" check expects 404 (one line).

4. Note: a failed cleanup leaves one orphan file (by design, and logged)

Covered by test E-03. If both the database record and the file deletion fail, the file stays in storage. Both `upload_cleanup_failed` and `upload_record_failed` are logged.

First report's defects, re-checked

First report	Status
e2e <code>getByRole("alert")</code> matched Next's route announcer	Fixed in 2265875: the locator is scoped to the upload form; the e2e suite passes 28/28
Unknown professor returns 400, not 404	Fixed on this branch: see Finding 1
The upload API was open to anyone	Fixed in 83f8b16: tests A-01 to A-06 confirm 403

Results per test

Each test lists the numbers of the calls it made. These numbers link to the call table.

A. POST /api/admin/uploads — access control

ID	Test	Result	Time	Calls
A-01	logged-out visitor gets 403 and nothing is stored	PASS	14 ms	1
A-02	logged-in student gets 403 and nothing is stored	PASS	3 ms	2
A-03	admin with an expired session gets 403	PASS	2 ms	3
A-04	unknown session token gets 403	PASS	2 ms	4
A-05	session lookup failure fails closed with 403	PASS	6 ms	5
A-06	a student's oversized request is 403, not 413 (auth is checked first)	PASS	2 ms	6

B. POST /api/admin/uploads — request handling

ID	Test	Result	Time	Calls
B-01	Content-Length over 20 MB + 1 MB overhead is refused with 413 before the body is read	PASS	1 ms	7
B-02	Content-Length exactly at the limit passes the size gate	PASS	2 ms	8
B-03	a JSON body is refused with 400 invalid_form	PASS	1 ms	9
B-05	a streamed body with no Content-Length stops being read at the limit and gets 413	PASS	4 ms	10
B-04	a non-numeric Content-Length header does not crash the route	PASS	7 ms	11

C. POST /api/admin/uploads — form fields

ID	Test	Result	Time	Calls
C-01	valid exam with every optional field returns 201 and the file link	PASS	10 ms	12
C-02	valid material with only required fields returns 201	PASS	4 ms	13
C-03	missing category is refused with 400	PASS	2 ms	14
C-04	lowercase category is refused with 400	PASS	2 ms	15
C-05	unknown category is refused with 400	PASS	2 ms	16
C-06	missing course is refused with 400	PASS	2 ms	17
C-07	missing title is refused with 400	PASS	1 ms	18
C-08	whitespace-only title is refused with 400	PASS	1 ms	19
C-09	title of 151 characters is refused with 400	PASS	1 ms	20
C-10	year 1949 is refused with 400	PASS	1 ms	21
C-11	two-digit year is refused with 400	PASS	1 ms	22
C-12	year two years ahead is refused with 400	PASS	1 ms	23
C-13	decimal year is refused with 400	PASS	1 ms	24
C-14	topic of 101 characters is refused with 400	PASS	2 ms	25
C-15	lowercase exam type is refused with 400	PASS	1 ms	26
C-16	exam type on a material is refused with 400	PASS	1 ms	27
C-17	unknown term is refused with 400	PASS	1 ms	28
C-18	professor from another course is refused with 404, as the Sprint 1 plan says	PASS	1 ms	29

ID	Test	Result	Time	Calls
C-19	professor that does not exist is refused with 404, as the Sprint 1 plan says	PASS	1 ms	30
C-20	unknown course is refused with 404	PASS	2 ms	31
C-21	title of exactly 150 characters is accepted	PASS	5 ms	32
C-22	year 1950 is accepted	PASS	5 ms	33
C-23	next year is accepted	PASS	4 ms	34
C-24	topic of exactly 100 characters is accepted	PASS	7 ms	35
C-25	padded fields are trimmed is accepted	PASS	3 ms	36
C-26	file sent as a text field is refused with 400	PASS	2 ms	37
C-27	missing file is refused with 400	PASS	1 ms	38
C-28	rejection logs never contain the title, file name or a long course ID	PASS	2 ms	39

D. POST /api/admin/uploads — file checks

ID	Test	Result	Time	Calls
D-01	.exe is refused	PASS	1 ms	40
D-02	no extension is refused	PASS	1 ms	41
D-03	double extension .pdf.exe is refused	PASS	1 ms	42
D-04	.html is refused	PASS	2 ms	43
D-05	empty .pdf is refused	PASS	2 ms	44
D-06	HTML renamed to .pdf is refused	PASS	2 ms	45
D-07	PDF renamed to .docx is refused	PASS	2 ms	46
D-08	JPEG renamed to .png is refused	PASS	1 ms	47
D-09	truncated PDF header is refused	PASS	2 ms	48
D-10	a file over the request limit without a Content-Length header is refused with 413	PASS	68 ms	49
D-11	a file of exactly 20 MB is accepted	PASS	145 ms	50
D-12	the browser's MIME type is ignored; the stored type comes from the checked extension	PASS	6 ms	51
D-13	an uppercase extension is accepted and stored with a lowercase key	PASS	6 ms	52
D-14	Unix path traversal in the file name cannot escape the storage folder	PASS	8 ms	53
D-15	Windows path traversal in the file name cannot escape the storage folder	PASS	10 ms	54
D-16	a plain ZIP renamed to .docx is refused as corrupt	PASS	3 ms	55
D-17	a DOCX renamed to .pptx is refused as corrupt	PASS	2 ms	56

E. POST /api/admin/uploads — storage and record failures

ID	Test	Result	Time	Calls
E-01	a storage write failure returns 500, logs once with the error code, and creates no record	PASS	3 ms	57
E-02	a database record failure returns 500 and deletes the stored file	PASS	4 ms	58

ID	Test	Result	Time	Calls
E-03	when cleanup also fails, both failures are logged (the orphan file stays)	PASS	5 ms	59
E-04	an unexpected database error returns a generic 500 without leaking details	PASS	2 ms	60
E-05	two uploads with the same file name get different storage keys and both survive	PASS	7 ms	61, 62
E-06	the returned link URL-encodes the file ID	PASS	5 ms	63

F. Upload then GET /files/:fileId — file retrievable after upload

ID	Test	Result	Time	Calls
F-01	PDF exam opens byte-for-byte with the right headers	PASS	8 ms	64, 65
F-02	DOCX material opens byte-for-byte with the right headers	PASS	6 ms	66, 67
F-03	PPTX material opens byte-for-byte with the right headers	PASS	6 ms	68, 69
F-04	PNG exam opens byte-for-byte with the right headers	PASS	6 ms	70, 71
F-05	JPG exam opens byte-for-byte with the right headers	PASS	6 ms	72, 73
F-06	a refused upload leaves no file to open	PASS	3 ms	74, 75
F-07	a file deleted from storage after upload shows the 404 page, not an error	PASS	8 ms	76, 77

G. getUploadOptions — course list for the upload form

ID	Test	Result	Time	Calls
G-01	returns courses ordered by code with their professors flattened	PASS	13 ms	—
G-02	a database failure is logged and rethrown for the error page	PASS	4 ms	—

H. validateUploadFile and parseUploadForm — direct calls

ID	Test	Result	Time	Calls
H-01	an empty .exe is reported as unsupported (type is checked before size)	PASS	4 ms	—
H-02	a 2-byte JPEG is reported as corrupt, not crashed on	PASS	3 ms	—
H-03	parseUploadForm uses the given clock for the year limit	PASS	3 ms	—
H-04	parseUploadForm drops blank optional fields	PASS	8 ms	—

Results per API call

Every call the tests made, with the response the real code returned. Section F calls open the file that the call before them uploaded. Log events are the `logError` events emitted during the test up to and including that call. A call's **Result** is its test's result. The status is coloured green for 2xx, amber for 4xx and red for 5xx; amber and red are expected outcomes here.

#	Test	Request	Status	Response	Log events	Result
1	A-01	POST /api/admin/uploads [no session] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	403	{"reason":"forbidden","error":"Only an admin can upload files."}	unauthorized_access	PASS
2	A-02	POST /api/admin/uploads [student-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	403	{"reason":"forbidden","error":"Only an admin can upload files."}	unauthorized_access	PASS
3	A-03	POST /api/admin/uploads [expired-admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	403	{"reason":"forbidden","error":"Only an admin can upload files."}	session_validation_failed, unauthorized_access	PASS
4	A-04	POST /api/admin/uploads [forged-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	403	{"reason":"forbidden","error":"Only an admin can upload files."}	session_validation_failed, unauthorized_access	PASS
5	A-05	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	403	{"reason":"forbidden","error":"Only an admin can upload files."}	session_validation_failed, unauthorized_access	PASS
6	A-06	POST /api/admin/uploads [student-token] headers={"content-length":"52428800"} body="x"	403	{"reason":"forbidden","error":"Only an admin can upload files."}	unauthorized_access	PASS
7	B-01	POST /api/admin/uploads [admin-token] headers={"content-length":"22020097"} body="x"	413	{"reason":"file_too_large","error":"The file is larger than 20 MB."}	upload_rejected	PASS
8	B-02	POST /api/admin/uploads [admin-token] headers={"content-length":"22020096"} body="x"	400	{"reason":"invalid_form","error":"The upload could not be read. Please try again."}	upload_rejected	PASS
9	B-03	POST /api/admin/uploads [admin-token] headers={"content-type":"application/json"} body="{\"title\":\"x\"}"	400	{"reason":"invalid_form","error":"The upload could not be read. Please try again."}	upload_rejected	PASS
10	B-05	POST /api/admin/uploads [admin-token] endless 1 MB chunks, no Content-Length	413	{"reason":"file_too_large","error":"The file is larger than 20 MB."}	upload_rejected	PASS
11	B-04	POST /api/admin/uploads [admin-token] headers={"content-length":"abc","content-type":"text/plain"} body="x"	400	{"reason":"invalid_form","error":"The upload could not be read. Please try again."}	upload_rejected	PASS
12	C-01	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" professorId="prof-a" termId="term-fall" year="2025" examType="FINAL" topic="Routing" file=<file final.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}	—	PASS
13	C-02	POST /api/admin/uploads [admin-token] category="MATERIAL" courseId="course-a" title="Lecture 1" file=<file lecture1.pptx, 44 B, type "">	201	{"file":{"id":"file-1","title":"Lecture 1","category":"MATERIAL","courseId":"course-a","url":"/files/file-1"}}	—	PASS

#	Test	Request	Status	Response	Log events	Result
14	C-03	POST /api/admin/uploads [admin-token] category="" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Choose whether this is an exam or a material."}	upload_rejected	PASS
15	C-04	POST /api/admin/uploads [admin-token] category="exam" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Choose whether this is an exam or a material."}	upload_rejected	PASS
16	C-05	POST /api/admin/uploads [admin-token] category="HOMEWORK" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Choose whether this is an exam or a material."}	upload_rejected	PASS
17	C-06	POST /api/admin/uploads [admin-token] category="EXAM" courseId="" title="Final Exam" file=<file final.pdf, 27 B, type "">	400	{"reason":"course_required","error":"Choose a course."}	upload_rejected	PASS
18	C-07	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Enter a title."}	upload_rejected	PASS
19	C-08	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title=" " file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Enter a title."}	upload_rejected	PASS
20	C-09	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="tttttttttttttttttttttt...(151 chars) file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_fields","error":"Title must be 150 characters or fewer."}	upload_rejected	PASS
21	C-10	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="1949" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_year","error":"Year must be a 4-digit year between 1950 and 2027."}	upload_rejected	PASS
22	C-11	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="25" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_year","error":"Year must be a 4-digit year between 1950 and 2027."}	upload_rejected	PASS
23	C-12	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="2028" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_year","error":"Year must be a 4-digit year between 1950 and 2027."}	upload_rejected	PASS
24	C-13	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="2025.0" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_year","error":"Year must be a 4-digit year between 1950 and 2027."}	upload_rejected	PASS
25	C-14	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" topic="xxxxxxxxxxxxxxxxxxxxxxxx...(101 chars) file=<file final.pdf, 27 B, type "">	400	{"reason":"topic_too_long","error":"Topic must be 100 characters or fewer."}	upload_rejected	PASS
26	C-15	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" examType="final" file=<file final.pdf, 27 B, type "">	400	{"reason":"invalid_type","error":"Choose Midterm, Final, Quiz, or Other as the type."}	upload_rejected	PASS
27	C-16	POST /api/admin/uploads [admin-token] category="MATERIAL" courseId="course-a" title="Final Exam" examType="FINAL" file=<file final.pdf, 27 B, type "">	400	{"reason":"type_not_allowed","error":"Only exams have a type."}	upload_rejected	PASS
28	C-17	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" termId="term-nope" file=<file final.pdf, 27 B, type "">	400	{"reason":"term_not_found","error":"The selected term doesn't exist."}	upload_rejected	PASS

#	Test	Request	Status	Response	Log events	Result
29	C-18	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" professorId= prof-b" file=<file final.pdf, 27 B, type "">	404	{"reason": "professor_not_assigned", " error": "The selected professor doesn't teach the selected course."}	upload_rejected	PASS
30	C-19	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" professorId=" prof-ghost" file=<file final.pdf, 27 B, type "">	404	{"reason": "professor_not_assigned", " error": "The selected professor doesn't teach the selected course."}	upload_rejected	PASS
31	C-20	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-ghost" title="Final Exam" file=<file final.pdf, 27 B, type "">	404	{"reason": "course_not_found", "error": "The selected course doesn't exist."}	upload_rejected	PASS
32	C-21	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="ttttttttttttttttttttt...(150 chars) file=<file final.pdf, 27 B, type "">	201	{ "file": {"id": "file-1", "title": " tttttttttttttttttttttttttttttttttttttt tttttttttttttttttttttttttttttttttttttt tttttttttttttttttttttttttttttttttttttt tttttt..."} }	-	PASS
33	C-22	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="1950" file= <file final.pdf, 27 B, type "">	201	{ "file": {"id": "file-1", "title": "Final Exam", "category": "EXAM", "courseId": " course-a", "url": "/files/file-1"}} }	-	PASS
34	C-23	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" year="2027" file= <file final.pdf, 27 B, type "">	201	{ "file": {"id": "file-1", "title": "Final Exam", "category": "EXAM", "courseId": " course-a", "url": "/files/file-1"}} }	-	PASS
35	C-24	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" topic=" xxxxxxxxxxxxxxxxxxxxx...(100 chars) file=<file final.pdf, 27 B, type "">	201	{ "file": {"id": "file-1", "title": "Final Exam", "category": "EXAM", "courseId": " course-a", "url": "/files/file-1"}} }	-	PASS
36	C-25	POST /api/admin/uploads [admin-token] category="EXAM" courseId=" course-a " title=" Midterm " year=" 2024 " file=<file final.pdf, 27 B, type "">	201	{ "file": {"id": "file-1", "title": "Midterm", "category": "EXAM", "courseId": "course-a", " url": "/files/file-1"}} }	-	PASS
37	C-26	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file="not-a-file"	400	{ "reason": "invalid_fields", "error": " Choose a file to upload."} }	upload_rejected	PASS
38	C-27	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam"	400	{ "reason": "invalid_fields", "error": " Choose a file to upload."} }	upload_rejected	PASS
39	C-28	POST /api/admin/uploads [admin-token] category="EXAM" courseId="cccccccccccccccccc...(500 chars) title=" SECRET-TITLE" file=<file SECRET-NAME.pdf, 27 B, type "">	404	{ "reason": "course_not_found", "error": "The selected course doesn't exist."}	upload_rejected	PASS
40	D-01	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file setup. exe, 27 B, type "">	415	{ "reason": "unsupported_type", "error": " Unsupported file type. Upload a PDF, DOCX, PPTX, PNG, or JPG file."}	upload_rejected	PASS
41	D-02	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file README, 27 B, type "">	415	{ "reason": "unsupported_type", "error": " Unsupported file type. Upload a PDF, DOCX, PPTX, PNG, or JPG file."}	upload_rejected	PASS
42	D-03	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file exam. pdf.exe, 27 B, type "">	415	{ "reason": "unsupported_type", "error": " Unsupported file type. Upload a PDF, DOCX, PPTX, PNG, or JPG file."}	upload_rejected	PASS
43	D-04	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file page. html, 13 B, type "">	415	{ "reason": "unsupported_type", "error": " Unsupported file type. Upload a PDF, DOCX, PPTX, PNG, or JPG file."}	upload_rejected	PASS

#	Test	Request	Status	Response	Log events	Result
44	D-05	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file empty.pdf, 0 B, type "">	400	{"reason":"empty_file","error":"The selected file is empty."}	upload_rejected	PASS
45	D-06	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file fake.pdf, 38 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .pdf file."}	upload_rejected	PASS
46	D-07	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file fake.docx, 27 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .docx file."}	upload_rejected	PASS
47	D-08	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file fake.png, 6 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .png file."}	upload_rejected	PASS
48	D-09	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file short.pdf, 3 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .pdf file."}	upload_rejected	PASS
49	D-10	POST /api/admin/uploads [admin-token] multipart bytes with file=<file big.pdf, 22020097 B>, no Content-Length	413	{"reason":"file_too_large","error":"The file is larger than 20 MB."}	upload_rejected	PASS
50	D-11	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file max.pdf, 20971520 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
51	D-12	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file exam.pdf, 27 B, type "text/html">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
52	D-13	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file EXAM.PDF, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
53	D-14	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file ../../etc/evil.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
54	D-15	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file ../../Windows/evil.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
55	D-16	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file archive-renamed.docx, 8 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .docx file."}	upload_rejected	PASS
56	D-17	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file notes-renamed.pptx, 41 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .pptx file."}	upload_rejected	PASS
57	E-01	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	500	{"reason":"storage_failed","error":"The file could not be stored. Please try again."}	upload_storage_failed	PASS
58	E-02	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	500	{"reason":"record_failed","error":"The file could not be saved. Please try again."}	upload_record_failed	PASS

#	Test	Request	Status	Response	Log events	Result
59	E-03	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	500	{"reason":"record_failed","error":"The file could not be saved. Please try again."}	upload_cleanup_failed, upload_record_failed	PASS
60	E-04	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	500	{"reason":"upload_failed","error":"The upload failed. Please try again."}	upload_failed	PASS
61	E-05	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file same.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
62	E-05	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file same.pdf, 29 B, type "">	201	{"file":{"id":"file-2","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-2"}}}	—	PASS
63	E-06	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	201	{"file":{"id":"file 9/x","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file%209%2Fx"}}}	—	PASS
64	F-01	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file exam.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
65	F-01	GET /files/file-1	200	<application/pdf, 27 B, inline; filename="exam.pdf">	—	PASS
66	F-02	POST /api/admin/uploads [admin-token] category="MATERIAL" courseId="course-a" title="Final Exam" file=<file notes.docx, 41 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"MATERIAL","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
67	F-02	GET /files/file-1	200	<application/vnd.openxmlformats-officedocument.wordprocessingml.document, 41 B, attachment; filename="notes.docx">	—	PASS
68	F-03	POST /api/admin/uploads [admin-token] category="MATERIAL" courseId="course-a" title="Final Exam" file=<file slides.pptx, 44 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"MATERIAL","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
69	F-03	GET /files/file-1	200	<application/vnd.openxmlformats-officedocument.presentationml.presentation, 44 B, attachment; filename="slides.pptx">	—	PASS
70	F-04	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file scan.png, 10 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
71	F-04	GET /files/file-1	200	<image/png, 10 B, attachment; filename="scan.png">	—	PASS
72	F-05	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file scan.jpg, 6 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	—	PASS
73	F-05	GET /files/file-1	200	<image/jpeg, 6 B, attachment; filename="scan.jpg">	—	PASS
74	F-06	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file fake.pdf, 6 B, type "">	400	{"reason":"content_mismatch","error":"The file is corrupt or is not a real .pdf file."}	upload_rejected	PASS

#	Test	Request	Status	Response	Log events	Result
75	F-06	GET /files/file-1	404	<html: File unavailable page>	upload_rejected, file_record_not_found	PASS
76	F-07	POST /api/admin/uploads [admin-token] category="EXAM" courseId="course-a" title="Final Exam" file=<file final.pdf, 27 B, type "">	201	{"file":{"id":"file-1","title":"Final Exam","category":"EXAM","courseId":"course-a","url":"/files/file-1"}}}	–	PASS
77	F-07	GET /files/file-1	404	<html: File unavailable page>	physical_file_not_found	PASS

How to reproduce

```
M5_CALL_LOG=test-results/m5-calls.json npx vitest run \  
  tests/unit/cross-test-member-5-uploads.test.ts \  
  --reporter=json --outputFile=test-results/m5-results.json  
node cross-test-reports/member-5-uploads/build-tex.mjs  
tectonic cross-test-reports/member-5-uploads/member-5-uploads-cross-test.tex
```